

Quiz 3: Operational Semantics

EN.601.426/626 Principles of Programming Languages – SP26

Name: _____

Suppose we have a (S)imple (IMP)erative language, SIMP, with the following syntax (v : variable names, c : constants including integers):

(Expression)	$e ::= v \mid c$	(terminals)
	$\mid e + e \mid e - e \mid e \times e$	(integer operations)
	$\mid e = e \mid e < e \mid e > e$	(bool operations)
(Statement)	$s ::= s ; s$	(sequential)
	$\mid v := e$	(assignment)
	$\mid \text{If } e \{ s \} \text{ Else } \{ s \}$	(conditional)

1. Please write out a statement computing “absolute” value in the SIMP language. Assume the input is stored in a variable named **in** and the output should be stored in a variable named **out**.

2. The state of the program is represented as a mapping σ from variable names to their integer values (e.g., $\sigma = \{\text{in} \mapsto -5, \text{out} \mapsto 5\}$ and $\sigma(\text{in}) = -5$). Assume the semantics of expressions is expressed $\langle e, \sigma \rangle \Rightarrow y$ where $y \in \mathbb{Z} \cup \mathbb{B}$ and that of statements is noted as $\langle s, \sigma \rangle \Downarrow \sigma'$. Fill in the rules for the assignment and if statements. Use $\sigma[v \mapsto y]$ to denote σ with the value of v overwritten with y .

$$\text{(Sequence)} \quad \frac{\langle s_1, \sigma \rangle \Downarrow \sigma' \quad \langle s_2, \sigma' \rangle \Downarrow \sigma''}{\langle s_1; s_2, \sigma \rangle \Downarrow \sigma''}$$

(Assign) _____

(If-True) _____ (If-False) _____

3. Write out the proof tree for the execution of your absolute value program on the input $\sigma = \{\text{in} \mapsto x\}$ where $x \in \mathbb{Z}$ and $x < 0$. The following rules for expressions may be useful:

$$\begin{array}{ll}
 (\text{Var}) \quad \frac{v \in \text{dom}(\sigma) \quad y = \sigma(v)}{\langle v, \sigma \rangle \Rightarrow y} & (\text{Const}) \quad \frac{c \in \mathbb{Z}}{\langle c, \sigma \rangle \Rightarrow c} \\
 (\text{Sub}) \quad \frac{\langle e_1, \sigma \rangle \Rightarrow n_1 \quad \langle e_2, \sigma \rangle \Rightarrow n_2}{\langle e_1 - e_2, \sigma \rangle \Rightarrow n_1 - n_2} & (\text{Lt-True}) \quad \frac{\langle e_1, \sigma \rangle \Rightarrow n_1 \quad \langle e_2, \sigma \rangle \Rightarrow n_2}{n_1 < n_2 \Longrightarrow \langle e_1 < e_2, \sigma \rangle \Rightarrow \text{true}}
 \end{array}$$

Please make sure that your proof tree is complete: the leaf nodes should be the axiom rules (e.g., Var and Const).

$$x < 0 \Longrightarrow \langle \quad , \{\text{in} \mapsto x\} \rangle \Downarrow \{\text{in} \mapsto x, \text{out} \mapsto -x\}$$